

# MIT OpenCourseWare

5.61

Physical Chemistry 5.61 — Physical Chemistry May 2026 **EXTREMELY HARD –****MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (4 points) Analyze the limitations of standard approaches to Quantum Chemistry when applied to edge cases in 5.61. Construct explicit counterexamples showing where intuitive reasoning fails.
2. (3 points) Construct an original proof-level problem involving Spectroscopy for 5.61 (Physical Chemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
3. (4 points) Prove the fundamental theorem concerning Statistical Mechanics in the context of 5.61 (Physical Chemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
4. (5 points) Prove the fundamental theorem concerning Computational Chemistry in the context of 5.61 (Physical Chemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
5. (4 points) Analyze the limitations of standard approaches to Molecular Dynamics when applied to edge cases in 5.61. Construct explicit counterexamples showing where intuitive reasoning fails.
6. (4 points) Construct an original proof-level problem involving Quantum Chemistry for 5.61 (Physical Chemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
7. (5 points) Construct an original proof-level problem involving Spectroscopy for 5.61 (Physical Chemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
8. (9 points) Analyze the limitations of standard approaches to Statistical Mechanics when applied to edge cases in 5.61. Construct explicit counterexamples showing where intuitive reasoning fails.

9. (2 points) Construct an original proof-level problem involving Computational Chemistry for 5.61 (Physical Chemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
10. (4 points) Prove the fundamental theorem concerning Molecular Dynamics in the context of 5.61 (Physical Chemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
11. (3 points) Prove the fundamental theorem concerning Quantum Chemistry in the context of 5.61 (Physical Chemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
12. (2 points) Prove the fundamental theorem concerning Spectroscopy in the context of 5.61 (Physical Chemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
13. (3 points) Analyze the limitations of standard approaches to Statistical Mechanics when applied to edge cases in 5.61. Construct explicit counterexamples showing where intuitive reasoning fails.
14. (5 points) Prove the fundamental theorem concerning Computational Chemistry in the context of 5.61 (Physical Chemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
15. (6 points) Construct an original proof-level problem involving Molecular Dynamics for 5.61 (Physical Chemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
16. (2 points) Construct an original proof-level problem involving Quantum Chemistry for 5.61 (Physical Chemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
17. (7 points) Prove the fundamental theorem concerning Spectroscopy in the context of 5.61 (Physical Chemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
18. (3 points) Analyze the limitations of standard approaches to Statistical Mechanics when applied to edge cases in 5.61. Construct explicit counterexamples showing where intuitive reasoning fails.
19. (4 points) Analyze the limitations of standard approaches to Computational Chemistry when applied to edge cases in 5.61. Construct explicit counterexamples showing where intuitive reasoning fails.

20. (4 points) Analyze the limitations of standard approaches to Molecular Dynamics when applied to edge cases in 5.61. Construct explicit counterexamples showing where intuitive reasoning fails.
21. (2 points) Construct an original proof-level problem involving Quantum Chemistry for 5.61 (Physical Chemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
22. (4 points) Prove the fundamental theorem concerning Spectroscopy in the context of 5.61 (Physical Chemistry). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
23. (2 points) Analyze the limitations of standard approaches to Statistical Mechanics when applied to edge cases in 5.61. Construct explicit counterexamples showing where intuitive reasoning fails.
24. (5 points) Construct an original proof-level problem involving Computational Chemistry for 5.61 (Physical Chemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
25. (4 points) Construct an original proof-level problem involving Molecular Dynamics for 5.61 (Physical Chemistry) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

## Solutions

### Solution 1:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Quantum Chemistry. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

### Solution 2:

Solution approach: First recall the definitions and key theorems related to Spectroscopy from 5.61. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

### Solution 3:

The proof follows from the definitions and properties established in 5.61. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

### Solution 4:

The proof follows from the definitions and properties established in 5.61. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

### Solution 5:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Molecular Dynamics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

### Solution 6:

Solution approach: First recall the definitions and key theorems related to Quantum Chemistry from 5.61. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

### Solution 7:

Solution approach: First recall the definitions and key theorems related to Spectroscopy from 5.61. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

### Solution 8:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Statistical Mechanics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

### Solution 9:

Solution approach: First recall the definitions and key theorems related to Computational Chemistry from 5.61. Then apply the appropriate techniques to construct a

rigorous argument. The solution must be self-contained and reference only the material from the course.

**Solution 10:**

The proof follows from the definitions and properties established in 5.61. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

**Solution 11:**

The proof follows from the definitions and properties established in 5.61. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

**Solution 12:**

The proof follows from the definitions and properties established in 5.61. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

**Solution 13:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of Statistical Mechanics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

**Solution 14:**

The proof follows from the definitions and properties established in 5.61. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

**Solution 15:**

Solution approach: First recall the definitions and key theorems related to Molecular Dynamics from 5.61. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

**Solution 16:**

Solution approach: First recall the definitions and key theorems related to Quantum Chemistry from 5.61. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

**Solution 17:**

The proof follows from the definitions and properties established in 5.61. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

**Solution 18:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of Statistical Mechanics. By constructing explicit counterexamples, we see that the usual

hypotheses cannot be weakened without loss of the theorem's conclusion.

**Solution 19:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of Computational Chemistry. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

**Solution 20:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of Molecular Dynamics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

**Solution 21:**

Solution approach: First recall the definitions and key theorems related to Quantum Chemistry from 5.61. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

**Solution 22:**

The proof follows from the definitions and properties established in 5.61. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

**Solution 23:**

The edge case analysis reveals the subtle assumptions underlying standard treatments of Statistical Mechanics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

**Solution 24:**

Solution approach: First recall the definitions and key theorems related to Computational Chemistry from 5.61. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

**Solution 25:**

Solution approach: First recall the definitions and key theorems related to Molecular Dynamics from 5.61. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.